

SMART INDIA HACKATHON 2026

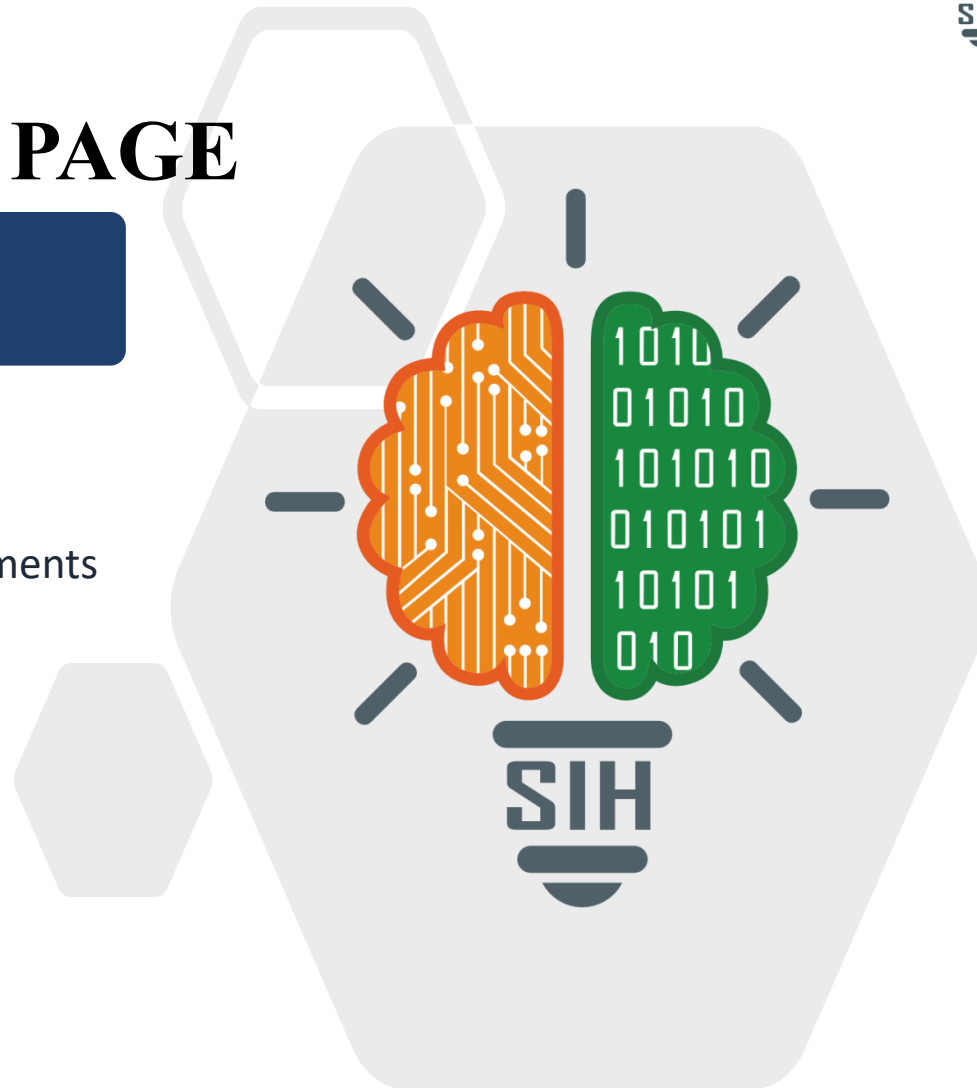


TITLE PAGE

TULA Unified Digital Lifecycle Platform for Legal Metrology

- **Problem Statement ID** – 26036
- **Problem Statement Title** – Development of an Online Verification System for Weighing and Measuring Instruments
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – _____
- **Team Name** – FriendlyFire

One instrument → one digital ID → one certificate anyone can check





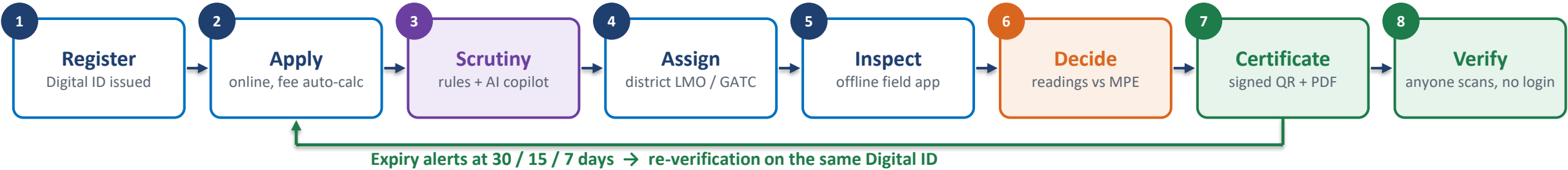
TULA: ONE INSTRUMENT, ONE DIGITAL ID



PROPOSED SOLUTION

A web + mobile platform where every weighing and measuring instrument gets a permanent Digital ID. Application, scrutiny, inspection, certificate, re-verification and enforcement all attach to that one ID.

DETAILED EXPLANATION: the life of one instrument in TULA



HOW IT ADDRESSES THE PROBLEM

Records on paper or isolated local systems
→ One central record per instrument

Manual scheduling and slow pendency
→ Rule-based fee, routing and live queues

Consumers can't check a stamp
→ Scan the QR, see validity instantly

Re-verification dates get missed
→ Automatic reminders before expiry

INNOVATION AND UNIQUENESS

Instrument-centric
One ID carries the full history across owners and years

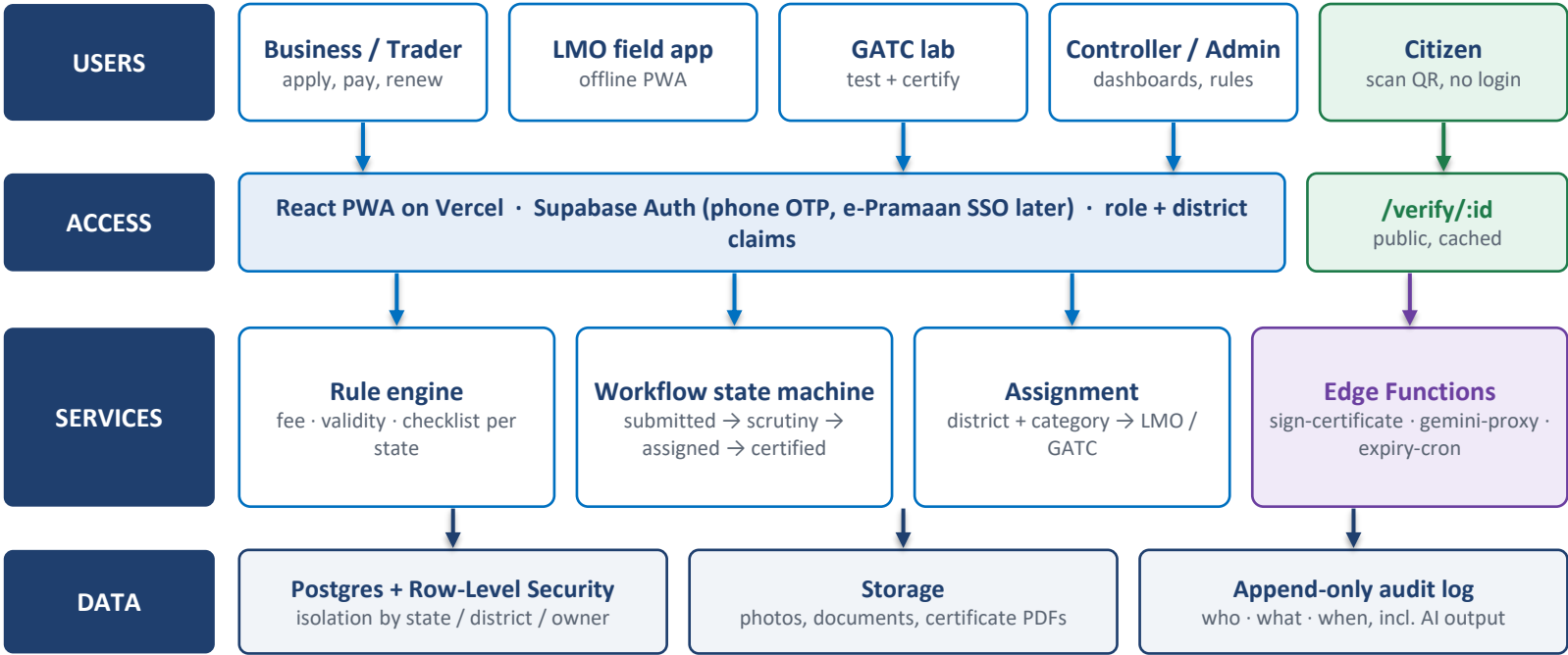
Signed QR
Ed25519 signature: a copied or edited certificate fails the scan

Rules as data
Each state sets fees, validity and checklists without code changes

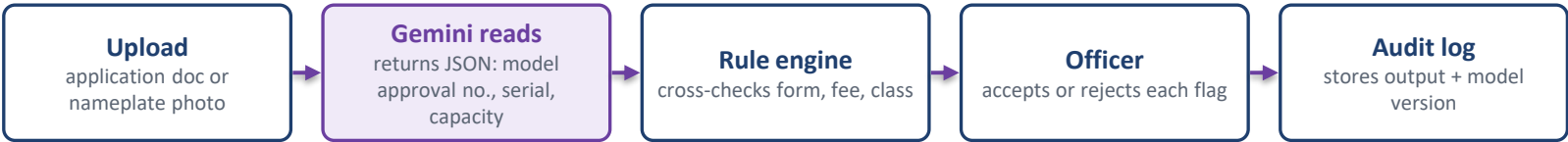
Gemini copilot
Reads documents and nameplates, flags mismatches. Officer decides

Citizen as auditor
Any shopper can scan and report a bad scale in two taps

METHODOLOGY AND PROCESS FOR IMPLEMENTATION (architecture)



WHERE GEMINI FITS: advisory only, the officer always decides



Also: help chat in Hindi / Gujarati / English grounded on the Act and Rules · citizen complaint triage · plain-language pendency summaries for Controllers.
Gemini never computes fees, MPE or pass/fail.

TECHNOLOGIES TO BE USED

- React 19 · TypeScript · Vite · Tailwind** role-based UI, installable PWA
- Workbox service worker · IndexedDB** offline inspections, sync on reconnect
- Supabase Postgres + RLS** one database, isolated per state / district
- Supabase Auth · Storage · Realtime · pg_cron** OTP login, files, live dashboards, daily expiry job
- Supabase Edge Functions** keys stay server-side, signing, Gemini proxy
- Google Gemini (Flash)** document + photo extraction, multilingual help
- Ed25519 + SHA-256 + QR** certificate verifiable even offline
- jsPDF · Recharts · Vercel** certificate PDF, reports, hosting

IMPLEMENTATION PHASES

- 1 Prototype (live): 6 roles, full lifecycle
- 2 Supabase backend, real auth, RLS
- 3 Signed QR, offline PWA, Gemini copilot
- 4 One-district pilot → state rollout



ANALYSIS OF THE FEASIBILITY OF THE IDEA: PS requirements vs build

PS requirement	Live prototype today	Pilot build
Stakeholder registration & profiles	🕒 6 demo roles, profile view	Phone OTP, e-Pramaan SSO
Apply: verification / re-verification	✓ 4 service types, auto fee	Treasury / PayGov payment
Scheduling & allocation to LMO / GATC	✓ slots, LMO or GATC route	District-aware auto-assign
Inspection observations recorded	✓ checklist + MPE table	Readings mandatory, auto verdict
QR digital certificate	✓ PDF + QR + SHA-256	Ed25519-signed, any device
Validity tracking & expiry alerts	✓ status + in-app alerts	Daily job → SMS / email
Dashboards: status, pendency, enforcement	✓ per-role dashboards	Realtime, state-wide
Photo & document upload	🕒 simulated attachments	Supabase Storage + Gemini
Export / print certificates & reports	✓ PDF, print, CSV	Bulk export
Mobile app for field officers	🕒 web, GPS, camera	Installable offline PWA

✓ working 🕒 partial in prototype, completed in pilot build

Technical
Open-source stack.
Prototype already runs the full lifecycle

Operational
Mirrors today's LMO / GATC process. No new roles

Economic
Free tiers cover a one-district pilot. No licence fees

Legal
LM Act 2009, General Rules 2011, GATC Rules 2013

POTENTIAL CHALLENGES AND RISKS → STRATEGIES

Weak network at field sites	Offline queue in the PWA. Signed QR verifies without internet
Fees and validity differ by state	Rules stored as data per state. No redeploy
Forged or copied certificates	Ed25519 signature, public scan, revocation list
AI misreads a document	Gemini only suggests. Officer confirms. Every output logged
Personal data (DPDP Act 2023)	RLS, minimal public fields, keys never in the browser
Officer adoption	Few-tap mobile flow, local languages, one-district pilot
Payment / SMS outage	Manual receipt entry + in-app alerts as fallback



Try the live prototype
Scan to open TULA. Pick any of the 6 roles and run apply → inspect → certificate → verify.
tula-legal-metrology.vercel.app



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

LMOs & GATCs

- Daily queue sorted by district
- Readings and photos on the phone
- Certificate issued on site

Businesses

- Apply and pay without a visit
- Live status of every application
- Reminder before a stamp expires

Consumers

- Scan any scale or pump QR
- See if it is valid right now
- Report a problem in two taps

Government

- Pendency by district, live
- Every action in one audit trail
- Enforcement cases tracked to closure

PILOT KPIs: targets we will measure against today's baseline

≤ 7 days

application → certificate (median)

< 3 sec

certificate check by QR scan

100%

due instruments reminded 30 days early

0

paper certificates issued in pilot district

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

Social

Fair weight at ration shops, fuel pumps and mandis. Shoppers can check for themselves.

Economic

Less pendency, fewer office visits and faster renewals for traders and MSMEs.

Environmental

Paperless applications, certificates and records. Fewer trips to the office.

Governance

Transparent, auditable trail from application to enforcement for every instrument.



RESEARCH AND REFERENCES



DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

RESEARCH FINDING	DESIGN DECISION IN TULA
1 Sec. 24 of the Act needs verification and stamping. GATC Rules 2013 add approved test centres as a second route.	One workflow for LMO and GATC, supervised by the Controller
2 Fees and validity periods differ across State Enforcement Rules.	Fee, validity and checklist stored as data per state + RLS
3 Instruments are re-verified every 1–2 years for their whole working life.	Permanent Digital ID with full history and expiry alerts
4 MPE limits come from OIML R 76 (scales) and R 117 (fuel dispensers), adopted in the Rules.	Pass / fail computed from recorded readings vs class MPE
5 A shopper cannot tell a genuine stamp from a fake one.	Signed QR anyone can scan without login, plus complaint button

References & Links

1. The Legal Metrology Act, 2009 · consumeraffairs.gov.in/pages/legal-metrology-act
2. Legal Metrology (General) Rules, 2011 & amendments
3. Legal Metrology (Government Approved Test Centre) Rules, 2013 & amendments
4. State Legal Metrology (Enforcement) Rules
5. OIML R 76-1 Non-automatic weighing instruments · oiml.org
6. OIML R 117-1 Dynamic measuring systems for liquids other than water
7. IT Act, 2000 & DPDP Act, 2023 electronic records, signatures, personal data
8. Supabase docs Row Level Security, Edge Functions · supabase.com/docs
9. Google Gemini API structured JSON output · ai.google.dev
10. Working prototype tula-legal-metrology.vercel.app